SAME PROCESS, DIFFERENT RISKS

Radical equations are solved by raising both sides to the index. Whether that step can create false answers depends on whether the index is odd or even.

RADICAL EQUATION

$$\sqrt[n]{f(x)} \quad g(x)$$

split by **whether the index n is odd or even**

CASE 1

odd index

Extraneous solutions are rare

negative radicands are handled naturally; cubing preserves sign

E.G. $\sqrt[3]{x-2} \quad 4 \Longrightarrow x = 66$

CASE 2

even index

Extraneous solutions can arise

fourth, sixth and other even roots behave like square roots: domain restrictions apply

TAKEAWAY

Check every candidate in the original equation when the index is even; see properties for why odd roots handle negative radicands.